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New paleomagnetic constraints on syntectonic growth strata in the western Qaidam Basin, NE Tibetan Plateau

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ABSTRACT

Accompanied by crustal shortening and mountain uplift caused by the large-magnitude south-north directed shortening under the collision between India and Eurasia, syntectonic growth strata in the western Qaidam Basin document important information about intense intracontinental deformation of the NE Tibetan Plateau. To better address the development of the Qaidam basin and the propagation of the growth strata associated with multiple tectonic uplifts of the Tibetan Plateau, we conducted a detailed paleomagnetic study on the late-Neogene fluvio-lacustrine sedimentary sequence exposed at two outcrop sections and a drill core in the western Qaidam Basin, and integrated the results with previous magnetostratigraphic studies. Based on well- determined ages of the centers and limbs of the two anticlines, we found an episodic late Cenozoic crustal shortening in the center of the western Qaidam Basin, as evidenced by syntectonic offlap growth strata deposited beginning at ~3.2 Ma. Combined with data from sedimentary facies and seismic profiles, we suggest that offlap growth strata progressively became younger from southwest to northeast in the center of the western Qaidam Basin due to strong basinward propagation of the eastern Kunlun fault and quick short-lived pulses of multiple tectonic uplifts of the NE Tibetan Plateau.

1. Introduction

The strong south-north directed shortening caused by the India-Eurasia collision has resulted in a series of intracontinental deformation processes in the Cenozoic basins of the NE Tibetan Plateau (Molnar and Tapponnier, 1975; Molnar and England, 1990; Leech et al., 2005; Sun and Zhang, 2009; Wang et al., 2014; Zhang et al., 2016; Zhang et al., 2018). To accommodate Cenozoic lithospheric deformation, intra-basinal strata have been folded and offset by thrust faults, and even formed synthetical folds (Xia et al., 2001; Sun et al., 2005; Zhou et al., 2006; Yin et al., 2008a, 2008b; Zhang et al., 2013, 2014; Bao et al., 2017), which can provide information on the development of basin and tectonic uplift histories of surrounding mountains. Syntectonic growth strata, reflecting Cenozoic shortening in intramontane basins, have played a key role in accommodating Cenozoic crustal shortening and mountain building (Zhou et al., 2006; Sun and Zhang, 2009; Bao et al., 2017; Zhang et al., 2018). Continuous and complete sedimentary sequences as growth strata of the anticlinorium provide information on extensional and contractional deformation associated with multiple rapid uplifts of the Tibetan Plateau and India-Asia convergence. To better understand the tectonic deformation of the Tibetan Plateau since the India-Eurasia collision, the age information on deformation of Cenozoic strata is crucial. Over the past few decades, dating by magnetostratigraphic studies of strata in the Cenozoic basins only focused on the commencement and terminal stages of sedimentation, and changes of the sedimentation rates (Song et al., 2005; Lu and Xiong, 2009; Li et al., 2014; Zheng et al., 2015; Ji et al., 2017; Wang et al., 2017; Fang et al., 2003, 2019), which provided little knowledge on the accurate ages of basin deformation and pulses of surrounding mountain uplift.

The Qaidam Basin, the largest intermontane basin of the NE Tibetan Plateau (Fig. 1A), is a closed inland basin and contains up to 12,000 m of Cenozoic sedimentary deposits (Fang et al., 2007). Under the dynamic mechanism of tectonic compression and crustal shortening

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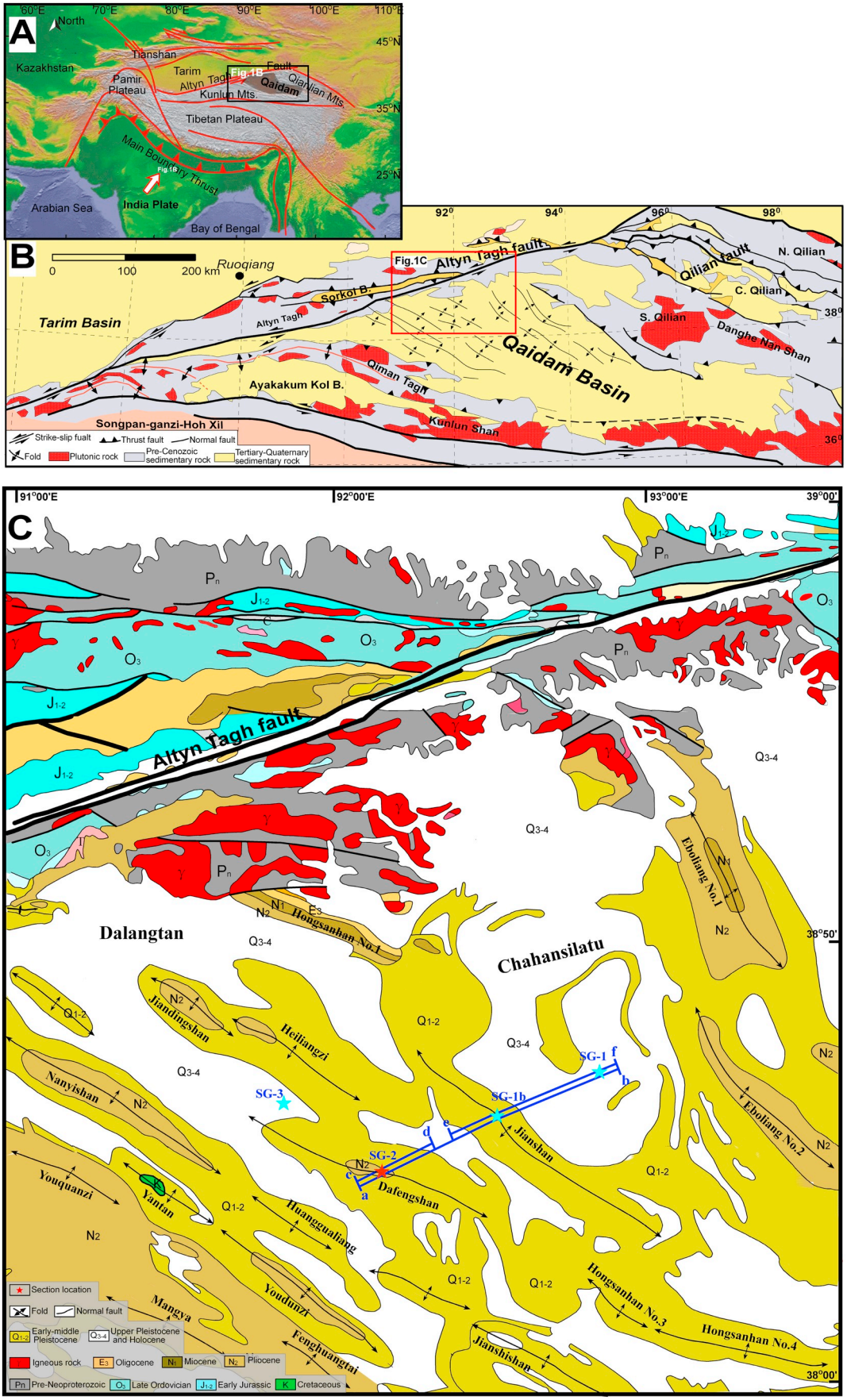


Fig. 1. (A) Tectonic DEM map of the Tibetan Plateau and it adjacent regions showing the major thrust faults (modified from Sun et al., 2005); (B) geologic map of the Qaidam basin and its adjacent regions within the northeastern Tibetan Plateau (modified from Yin et al., 2002); (C) geologic map of the study area in the western Qaidam Basin showing the locations of outcrop and drill core sections.

during India-Eurasia collision, the interior of the western Qaidam Basin has formed parallel sets of anticlines (Fig. 1B), extending for about 200 km from the southern edge of the Altyn Tagh mountains to the center of the basin (Fig. 1B). These thick sedimentary sequences and associated internal deformation characteristics make this area an ideal place to study the basin evolutionary history associated with major tectonics in the Tibetan Plateau. Three long continuous drill cores (SG-1, SG-1b, and SG-3) were recovered from the interior of the western Qaidam Basin to study the tectonic deformation history and the climatic evolution of the basin (Herb et al., 2013; Zhang et al., 2012a, 2012b; Cai et al., 2012; Wang et al., 2013; Yang et al., 2013; Lu et al., 2015). To better reveal the development of the basin and propagation of the growth strata associated with multiple pulses of deformation in the Tibetan Plateau, we carried out magnetostratigraphic studies on another drill core (SG-2) and two surface sections that comprise the complete post late Pliocene-early Pleistocene fluvio-lacustrine sedimentary sequence in the western Qaidam Basin. We then discuss the age of the deposition of syntectonic growth strata in the region and the potential tectonic driving mechanism for offlap growth strata.

2. Geologic setting

The Qaidam Basin, with an area of 121,000 km^2, is located in the northeastern Tibetan Plateau (Fig. 1A). The interior of the basin has an average elevation of about 3000 m and is surrounded by mountain ranges with average elevations of about 4000 to 5000 m, such as the Kunlun Shan (Mts.) to the south, the Qilian Shan (Mts.) to the northeast, and the Altyn Tagh Shan (Mts.) to the northwest (Fig. 1B). The North Qaidam thrust belt (NQMTB) is located in the southernmost part of the Cenozoic Qilian Shan-Nan Shan thrust belt (Yin et al., 2008a, 2008b; Li et al., 2017, 2018) (Fig. 1B). It consists of a few basinward thrust-folds (southwestwards), which expose the Paleozoic sedimentary rocks and formed the Yardang landform by Quaternary erosion (Wang et al., 2007; Kapp et al., 2011). The eastern Kunlun fault, derived from the Kunlun fault belt along the Qiman Tagh and the Kunlun Shan, bounds the southern margin of the Qaidam Basin (Fig. 1B). A NW-SE striking thrust-fold system, which consists of the southwestern Qaidam marginal thrust belt (SQMTB) (Yin et al., 2002; Wang et al., 2006; Fang et al., 2007) (Fig. 1B) and many sub-parallel basinward (northward) thrust-folds (Zhang et al., 2013, 2014), developed in the western Qaidam Basin since the early Cenozoic. The impressive NE-SW sinistral strike-slip Altyn Tagh fault cuts through the northwestern margin of the Qaidam Basin, resulting in displacements of ~350–700 km (Tapponnier et al., 1982; Peltzer et al., 1989; Ritts and Biffi, 2000; Meng et al., 2001; Chen et al., 2002). This large displacement has not only caused the formation of the Altyn Tagh mountains, but also led to the depression and deformation of the northwestern margin of the Qaidam Basin (Molnar and Tapponnier, 1975; Huang et al., 1996; Chen et al., 1999; Tapponnier et al., 2001; Yin et al., 2002, 2007, 2008a, 2008b; Wang and Burchfiel, 2004; Li et al., 2017) (Figs. 1b and 2).

The upper crust in the western Qaidam Basin during the Cenozoic was strongly shortened in a NE-SW direction caused by the northward compression of the eastern Kunlun fault belt (Tapponnier et al., 2001; Wang and Burchfiel, 2004; Zhou et al., 2006; Bao et al., 2017). As part of the crustal shortening and contractional deformation, the interior of the western Basin was deformed into multiple paralleled rows of anticlines (Yin et al., 2002; Wang et al., 2006; Zhang et al., 2014) (Fig. 1B, C). All the anticlines are visible in satellite images and characterized by linear NW-SE trending axes, indicating SW-NE crustal shortening. The anticlines in the western Qaidam Basin have been divided into two thrust-fold systems, the Kunlun and Qilian Shan-Nan Shan fault/thrust belt (Zhang et al., 2014). The former consists of the Kunbei, Alaer, Shizigou, Youshashan, Ganchaigou, Xianshuiquan, Youquanzi, Yantan, Youdunzi, Nanyishan, Huanggualiang, Xiaoliangshan, Dafengshan, Jiandingshan and Jianshan folds, and the latter comprises the Saishiteng Shan, Xitie Shan, Leng Hu, Nanbaxian, Eboliang and Yahu folds (Zhang et al., 2014) (Fig. 1). Accompanied by uplift of the surrounding mountains during the outward growth of the Tibetan Plateau since the Eocene (Wang and Coward, 1990; Métivier et al., 1998; Xia et al., 2001; Fang et al., 2007; Wang et al., 2008a, 2014; Yin et al., 2008b), shortening-related growth-strata are widely preserved within folds in the western part of the basin (Zhang et al., 2013, 2014; Zhang et al., 2018). Outcrop exposures and seismostratigraphic profiles show that these growth strata have a common feature with thin sediment sequences in the axial regions and thick sediment sequences in the limbs of folds (Zhang et al., 2014), which are possibly caused by uplifts of anticline hinges and shifts of the basin depocenter associated with compression along faults from the basin margin (Sobel et al., 2003). Growth-strata in the western Qaidam Basin started to develop in the Oligocene (Xia et al., 2001; Yin et al., 2002; Zhou et al., 2006; Wang et al., 2007; Liu et al., 2009), and then experienced strong contractional deformation due to rapid strike-slip displacement along the large-scale Qilian, Altyn and Kunlun faults since the Pliocene (Zhang et al., 2013, 2014). The interior of the western Qaidam Basin had undergone multiple tectonic uplifts during the Quaternary, triggering an eastward shift of the sediment depocenter and leading to extensive development of growth strata (Fang et al., 2007; Huang et al., 1996; Métivier et al., 1998; Kapp et al., 2011).

The western Qaidam Basin contains a 6000 m thick continuous and complete sequence of Cenozoic sedimentary deposits sourced from the surrounding mountains (Xia et al., 2001; Yin et al., 2002; Zhou et al., 2006; Wang et al., 2007; Liu et al., 2009), recording the interaction of rivers and lakes during depression and uplift of the basin (Wang and Burchfiel, 2004; Fang et al., 2007). The margins of deposits in the western Qaidam Basin are characterized by thick layers of sandstone, conglomerate and thin mudstone of Eocene age, while the basin center contains deposits of very thick mudstone, calcareous mudstone and marl, with intercalated gypsum and halite of Oligocene to late Pliocene age (Zhang et al., 2012a, 2012b). Since the late Pliocene, sedimentary deposits in the center of the Qaidam Basin are characterized by clay, mudstone, thick halite and gypsum, and saline lacustrine deposits (Wang and Burchfiel, 2004; Zhang et al., 2012b, 2014). The deposition of thick 200 m halite layers in the western Qaidam Basin terminated at different times at different localities, as revealed by sedimentary facies and paleomagnetic age estimates of cores recovered from boreholes (Shi et al., 2010; Wang et al., 2012, 2013; Zhang et al., 2012a, 2014; Cai et al., 2012). The sedimentary sequences from the late Miocene - late Pliocene Shizigou formation to the Quaternary Qigequan formation are well exposed in the center of the western Qaidam Basin (QBGM, 1991). Three boreholes (SG-1, SG-1b, and SG-3) were drilled at the limb and in the center of the Jianshan (JS) and the limb of the Dafengshan (DFS) anticlines (see locations in Fig. 1C), and were magnetostratigraphically dated to be about 2.77 Ma - 0.1 Ma, 7.3 Ma - 1.6 Ma, and 3.06 Ma - 0.1 Ma, respectively (Zhang et al., 2012a, 2014; Cai et al., 2012).

3. Studied sections and drill core

The 80 m-thick Dafengshan-I section (DFS-I) (38°12′32.7″N, 92°3′40.2″E) is located in the SW limb of the Dafenshan (DFS) anticline (Figs. 1C and 2). The overall section here consists mainly of yellow - gray fine clay and mudstone, intercalated with thin or scattered gypsum (Fig. 2B). Based on the lithology and sedimentary structures, the depositional environment for this section is identified as shallow brackish lake, which contains evidence for a lithofacies with mudflat environment under subaerial exposure (Phillips et al., 1993; Wang et al., 2013).

The Dafengshan-II section (DFS-II) (38°12′45.3″N, 92°3′32.9″E) has a thickness of 270 m, and is well exposed between the DFS-Isection and the SG-2 borehole (DFS) (Fig. 2B, C). Its sedimentary sequence is divided into three lithologic segments from top to bottom: (1) the upper part (0–52 m) mainly consists of light gray sandstone, gray conglomerate, white halite and gray mudstone with scattered gypsum,

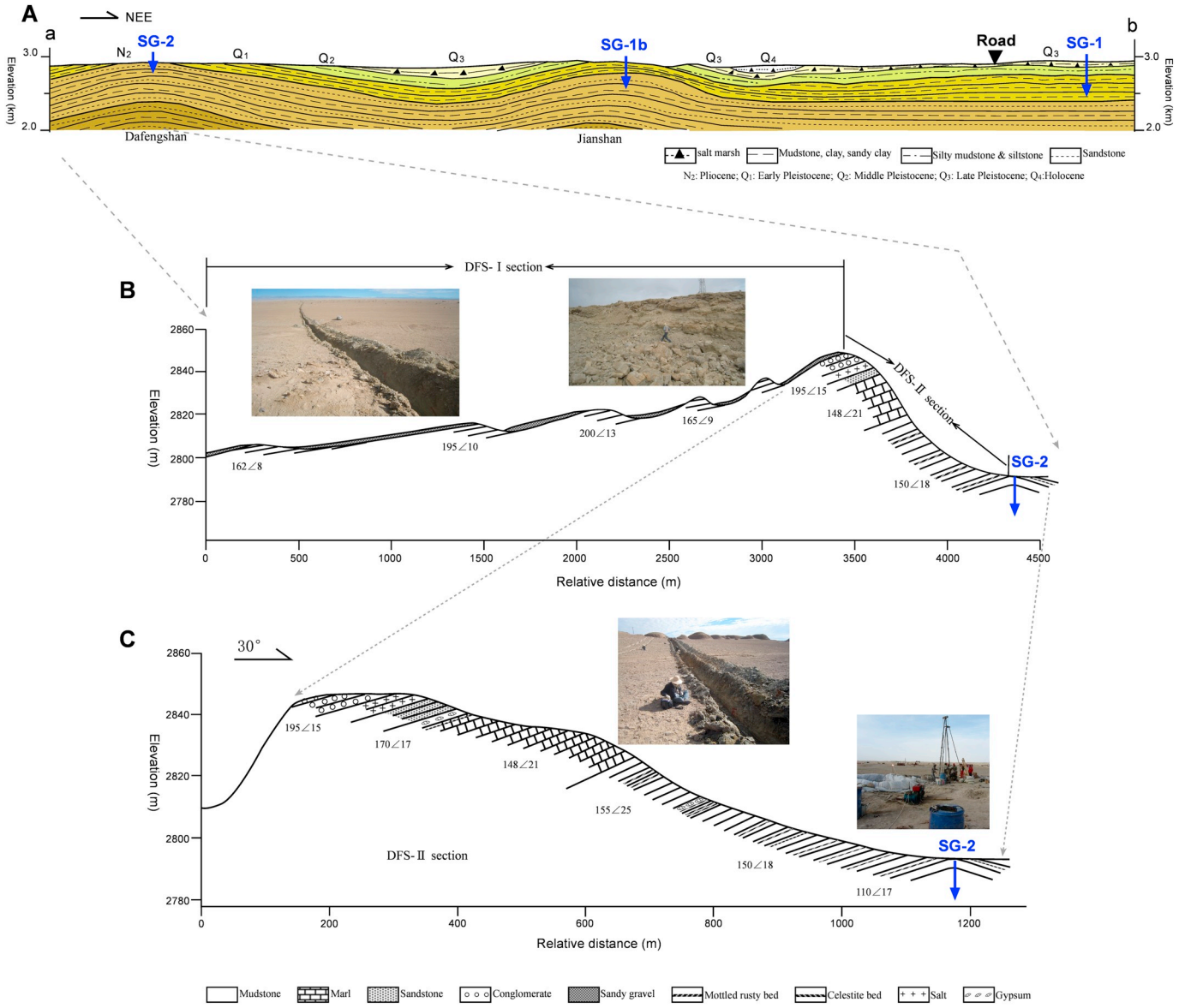


Fig. 2. (A) Schematic cross-section along a-b, including the SG-1, SG-1b and SG-2 drilling locations shown in Fig. 1C; (B) enlarged cross-section through the Dafengshan-I (DFS-I, with field photos) and Dafengshan-II (DFS-II) sections; (C) enlarged cross-section of the DFS-II section, with field photos.

suggesting a playa lake and fluvial environment; (2) the middle part (52–127 m) is composed of light gray marl and limestone, indicating a semi-deep lake under an alpine semiarid climate with moderate seasonality environment (Wang et al., 2012); and (3) the lower part (127–270 m) mainly comprises gray mudstone and calcareous mudstone, with scattered gypsum at depth of 168 m and one layer of celestite bed at depth of 192 m, reflecting a shallow semi-brackish lake environment.

The drill core SG-2 (38°13′4.2″N, 92°3′56.3″E) was drilled below the bottom of the DFS-II in the DFS anticline (Fig. 2B, C). It reached a depth of 151 m (core diameter 68 mm) with an average sedimentary recovery rate of about 90%. Its sedimentary sequence is dominated by dark gray clay, clay-silt, and calcareous mudstone, indicating a shallow-water, lacustrine environment. It is similar to the lower part of the DFS-II section.

The studied DFS-I and DFS-II sections and SG-2 drill cores are 501 m thick in total. The previous paleomagnetic studies of the SG-1, SG-1b and SG-3 drill cores (Zhang et al., 2012a, 2014; Cai et al., 2012) suggest that these sediments were deposited since the late Pliocene. As there is no unconformity in the whole sequence, we assume that the two new sections DFS-I and DFS-II and the drill core SG-2 comprise a complete and relative continuous late Pliocene - early Pleistocene sedimentary sequence in the western Qaidam Basin.

4. Sampling and measurements

The fresh rock exposures of the DFS-I and DFS-II sections are almost armored with thick indurated salt crust. We removed this cover by digging a 0.5–1 m deep trench to collect samples (Fig. 2B, C), and dug deep pits into bedrock exposures for the part without salt cover. To sample the lower part of the sedimentary sequence that was not exposed in the DFS sections, and the SG-2 drill core was recovered (Fig. 2C). A total of 326 oriented block samples were collected at 1 m stratigraphic intervals from the DFS-I and DFS-II surface sections, and then each block sample was cut into three sets of 2 × 2 × 2 cm cubic specimens in the laboratory. A total of 230 2-cm-long cubic samples were taken from the SG-2 drill core, with sample intervals of less than1.0 m.

To identify the magnetic mineralogy of the sedimentary materials, eighteen representative specimens were selected for systematic

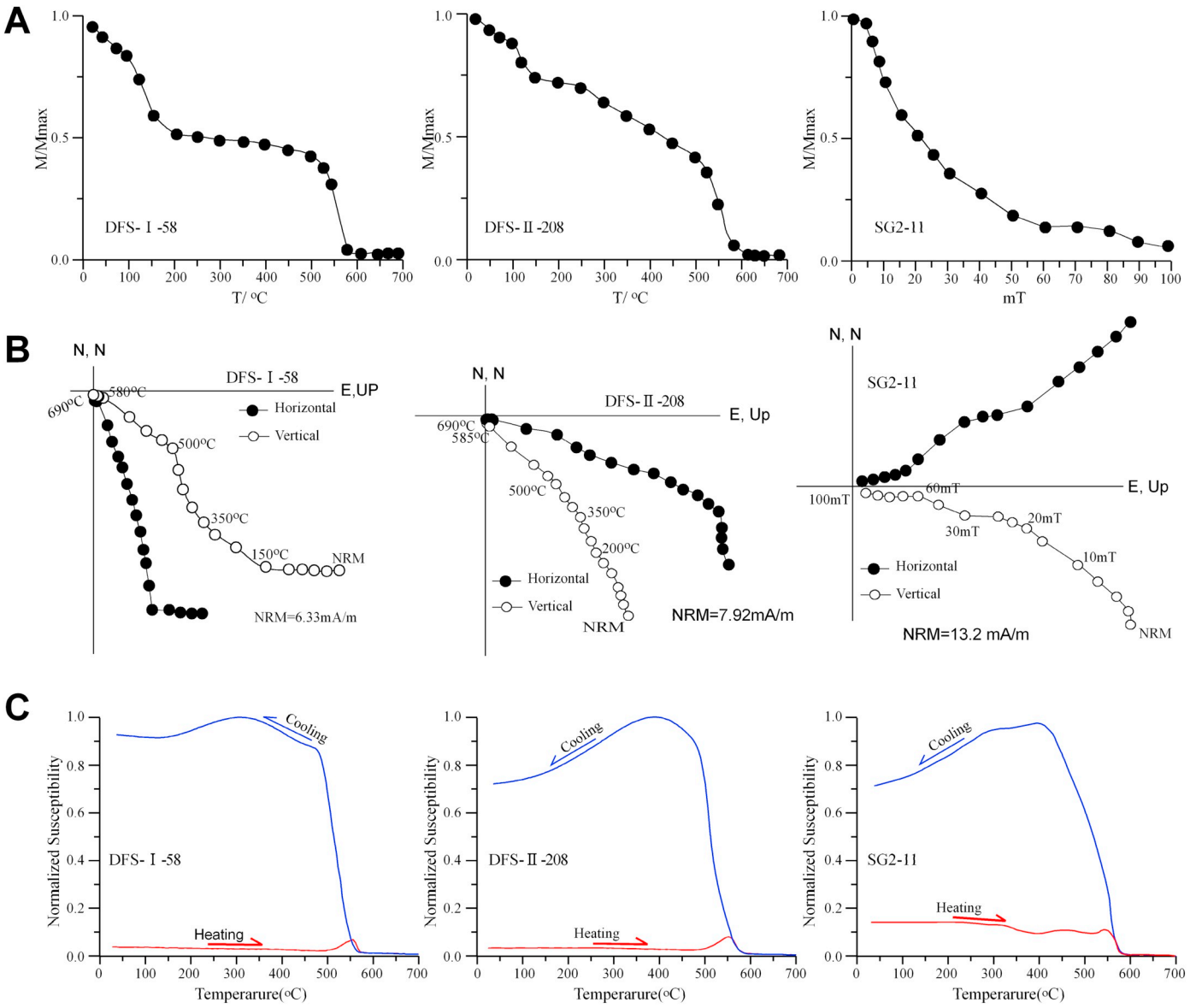


Fig. 3. Rock magnetic results of representative samples from strata exposed at the Dafengshan anticline: (A) Remanence intensity versus temperature and magnetic field from thermal and alternating field demagnetizations; (B) orthogonal vector diagrams - open/closed symbols represent vertical/horizontal projections; (C) κ-T curves showing magnetic susceptibility versus temperature (high temperature heating and cooling curves).

progressive thermal demagnetization and low-field thermomagnetic curves of magnetic susceptibility (κ-T curves). These specimens were gradually heated up to 585 °C in a TD-48 oven and measured in a 2G SQUID magnetometer (2G Enterprises). The κ-T curves were obtained between room temperature and ~700 °C on a MFK1-FA Kappabridge with a CS-4 high-temperature furnace (Agico Ltd., Brno, Czech Republic). Following the rock magnetic results (Fig. 3), all specimens were progressively alternating field (AF) demagnetized in 14 steps between 0 mT and 120 mT to obtain the characteristic remanent magnetization (ChRM) directions. AF magnetization was conducted by a 2G degausser attached to the magnetometer and remanence was measured using the superconducting rock magnetometer. Paleomagnetic samples of the two outcrop sections (DFS-I and DFS-II) were analyzed at the Institute of Tibetan Plateau Research, Chinese Academy of Sciences, and samples from the SG-2 drill core were measured at the Department of Geosciences, Tübingen University.

Representative thermal demagnetization diagrams show that a remanent magnetization component was readily removed below 150 °C (Fig. 3A), accompanied by a change of the remanence direction (Fig. 3B). Most samples showed a rapid decrease in remanence intensity above 500 °C (Fig. 3A), and then linearly decay toward the origin up to 585 °C (Fig. 3B). All the κ-T curves show a strong decrease in susceptibility from ~550–585 °C (Fig. 3C), indicating the presence of magnetite. Additional magnetite is formed during heating, as shown by the increase in χ in the heating curves from about 500 °C onward and much higher values in the cooling curves. However, the plateau-like curves before the start of the formation of new magnetite clearly indicate that magnetite is present in the original samples. There is also a decrease at about 300–400 °C, which could be related to destruction of maghemite. Based on thermal demagnetization and κ-T results, we suggest that magnetite is the main ChRM carrier. AF demagnetization shows that remanence intensities decrease to <5% of the NRM at 100 mT (Fig. 3A) and yield interpretable ChRM directions above 60 mT (Fig. 3B), further implying that a ferrimagnetic phase, like magnetite, is the dominant remanence carrier. The ChRM directions were estimated from at least five successive steps above 60 mT by principal component analysis (Kirschvink, 1980). Finally, 423 out of 526 specimens measured yielded reliable ChRM directions. The virtual geomagnetic

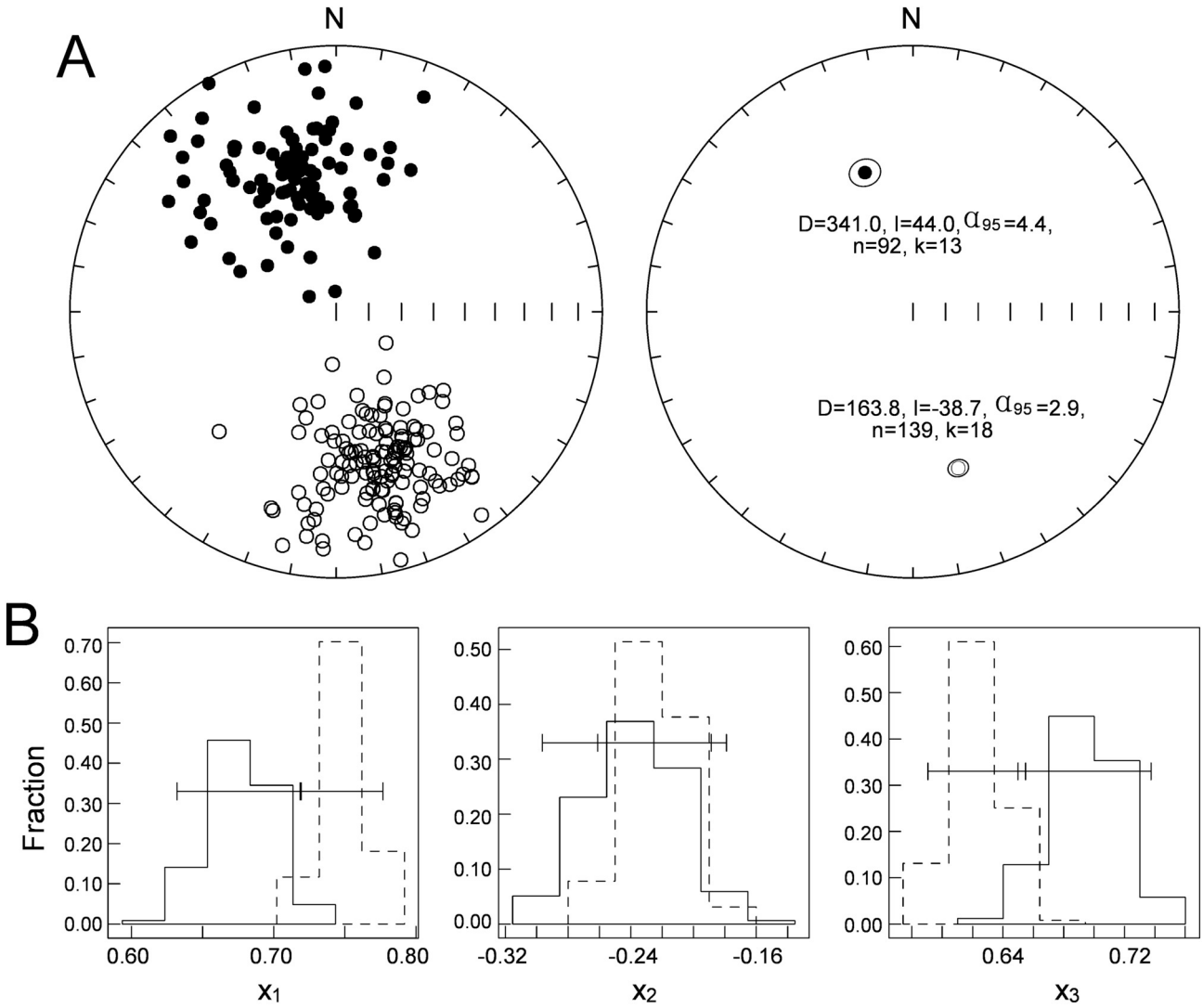


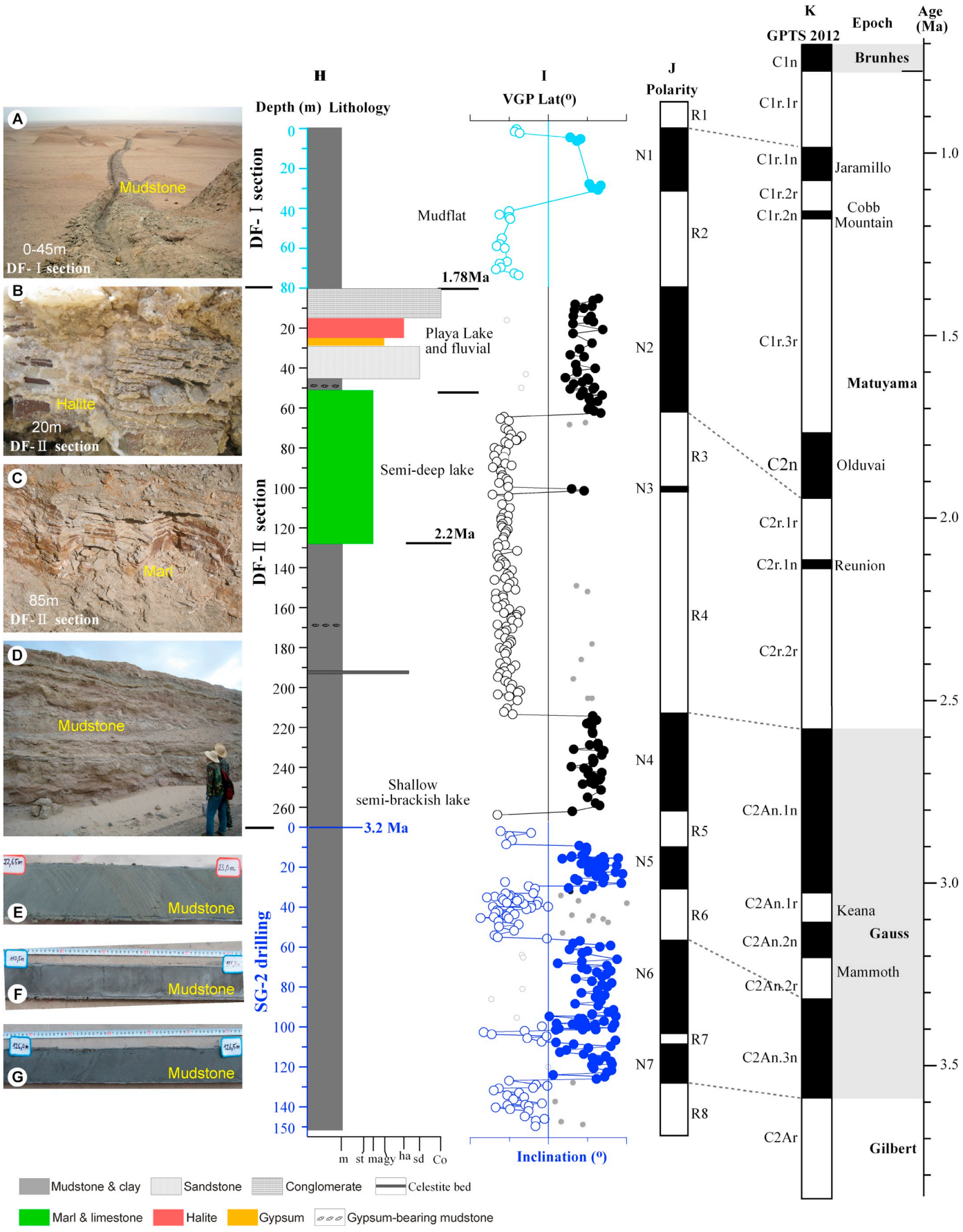
Fig. 4. (A) Equal-area projections of the ChRM directions and mean directions (with 95% confidence limits) for the DFS-I and DFS-II sections. α_{95}, N and k are the 95% confidence limit, the number of samples and the precision parameter, respectively. (B) Histograms of Cartesian coordinates of the bootstrap reversal test for the DFS-I and DFS-II sections. The intervals containing 95% of each set of components are shown (dashed and solid lines). The confidence bounds from the two data sets overlap in all three components, which suggest that the means of the reverse and normal modes cannot be distinguished at the 95% level of confidence, indicating a positive bootstrap reversal test (Tauxe, 1998).

polarity (VGP) latitudes from the DFS-I and DFS-II sections were calculated. Only inclination values could be extracted for the SG-2 borehole as there is no constraint on the core azimuth during the drilling.

5. Paleomagnetic results

The small variation in bedding dip values (8–21°) in the DFS-Iand DFS-II sections and the absence of declination data in the SG-2 drill core results (Fig. 2) do not allow applying a meaningful fold test, but a reversal test could be successfully performed on the ChRM data from the DFS-I and DFS-II sections. The tilt-corrected ChRM directions yield a mean normal polarity direction of D/I = 341.0°/+44.0° (k = 13.0, α_{95} = 4.4°, n = 118), and a mean reverse polarity direction of D/I = 163.8°/−38.7° (k = 18.0, α_{95} = 2.9°, n = 161) (Fig. 4A). The estimated means of the reverse and normal modes are indistinguishable at the 95% level of confidence, indicating a positive reversal test with classification B (Tauxe, 1998; McFadden and McElhinny, 1990). The Gaussian distribution of the ChRM vectors from a statistical bootstrap technique (Tauxe, 1998) demonstrates that the bootstrap reversal test is also positive (Fig. 4B). It implies that the secondary overprint magnetization was likely removed.

The VGP-latitudes along with the stratigraphic positions of the DFS-I and DFS-II sections and the inclination data from the SG-2 borehole were together applied to establish a composite magnetic polarity stratigraphy for the studied sequence. Each magneto-zone was defined by at least two successive data points of the same ChRM polarity, and a total of seven normal and eight reverse polarity magneto-zones, designated as N1-N7 and R1-R8, respectively, were identified (Fig. 5). The DFS-I and DFS-II sections include the Qigequan Formation, with a magnetostratigraphic age of <2.5 Ma (Zhang et al., 2013), as confirmed by paleontology and lithostratigraphy in the western Qaidam Basin (QBGM, 1991). Strata of this formation occur within the sequence from R1 - R4 and N1 - N4 in the upper part of our observed polarity zones (Fig. 5). The center of the DFS anticline contains the sedimentary sequence of the Shizigou Formation, indicating that R5 - R8 and N5 - N7 of the polarity zones from the SG-2 drill core correspond to late Miocene - late Pliocene (Zhang et al., 2013; Song et al., 2014). Based on the rough age constraints on these stratigraphic beds, the observed polarity zones from the DFS sections and the SG-2 drill core are well correlated with the chrons C1r.1r - C2Ar of the GPTS 2012 (Gradstein et al., 2012)



(caption on next page)

Fig. 5. Lithology and magnetostratigraphic composite results for the DFS-I and DFS-II sections and SG-2 drill core from the Dafengshan anticline in the western Qaidam Basin and their correlations with the GPTS 2012 (Gradstein et al., 2012). (A) Mudstone from the DFS-I section. (B-D) Halite, marl and mudstone from the DFS-II section. (E-G) Mudstone from the SG-2 drill core. (H) Lithologies for the DFS-I and DFS-II sections and SG-2 drill core. (I-K) The correlations between the observed polarity and the GPTS 2012. Note small gray open/closed circles represent single opposite samples within the multiple successive data points from one polarity zone.

(Fig. 5). The thick reverse magneto-zones R2, R3 and R4 can be directly correlated with the distinct polarity chrons C1r.3r, C2r.1r and C2r.2r within the Matuyama reverse polarity epoch, and the normal polarity zones N1, N2 and N3 can be readily correlated with the Jaramillo, Olduvai and Reunion normal polarity subchrons (events) (Fig. 5). The normal polarity zones N4, N5 and N6-N7 correspond to C2An.1n, C2An.2n and C2An.2n within the Gauss normal polarity epoch and the reverse polarity zones R5 and R6 likely represent the Kaena Event and Mammoth Event (Fig. 5). The top and bottom intervals R1 and R8 are considered to correlate with the lower part of chron C1r.1r from the Matuyama epoch and the upper part of chron C2Ar in the Gilbert epoch. Assuming these correlations with the GPTS to be realistic, the age of the sampled DFS-Iand DFS-II sections is estimated at 3.2 Ma to 0.9 Ma, and the SG-2 drill core has an age range of 3.7 Ma to 3.2 Ma (Fig. 5) based on extrapolation of the sedimentation rate between the uppermost/lowest two polarity boundaries.

6. Discussion

6.1. Implication of magnetic polarity stratigraphy for syntectonic growth strata

Understanding the depositional history of syntectonic growth strata, which are synchronous with the development and formation of a thrust fold belt, is crucial to interpreting the deformation of the stratigraphic sequence in the interior of the basin (Suppe et al., 1992; Anastasio et al., 1997; Sun and Zhang, 2009). Large magnitude clockwise block rotation, revealed by paleomagnetic studies (Dupont-Nivet et al., 2002; Chen et al., 2002; Sun et al., 2006; Li et al., 2017, 2018), influenced the sediment accommodation space and led to the formation of the array of en-echelon s-shaped structures in the western Qaidam Basin since the late Cenozoic (Wang et al., 2006; Zhang et al., 2014). The subsurface and well-exposed syntectonic growth strata in the western Qaidam Basin were deposited by limb rotation with clear changes of growth strata dip and thicknesses on the forelimb (Fig. 2) (Yin et al., 2007, 2008a, 2008b; Zhou et al., 2006; Liu et al., 2009; Wang et al., 2010; Zhang et al., 2018). This feature can also be implied by the ages of depositional termination of the sedimentary sequences exposed in the limbs and cores of anticlines.

Based on the proposed magnetostratigraphy of the SG-2 borehole strata, the youngest age of the center of the DFS anticline is 3.2 Ma (Figs. 5 and 6B), indicating that the tectonic event occurred at 3.2 Ma was one of the multiple deformations of syntectonic growth strata in the western Qaidam Basin. In contrast, the inferred paleomagnetic age is $<$0.9 Ma for the top of the DFS-II section (Figs. 5 and 6B), suggesting that the limb of the DFS stopped receiving sediments after about 0.9 Ma, as also indicated by the magnetostratigraphy of the SG-3 borehole in the NW limb of the DFS anticline that ceased deposition at ~0.1 Ma (Cai et al., 2012). The NW-SE directed Jianshan (JS) anticline (see location in Figs. 1B and 6A), which is parallel to the DFS anticline, also shows that the termination of sedimentation in the limb is younger than that in the center of the anticline. The youngest ages of the SG-1b drill core in the center of the JS anticline, and the SG-1 drill core in the adjacent sub-basin along the limb are estimated to be 1.6 Ma (Fig. 6D) (Zhang et al., 2012a) and 0.1 Ma (Fig. 6E) (Zhang et al., 2014), respectively. It indicates gradual younging of the surface strata from the hinge of the fold to the limb of the anticline (Zhang et al., 2014). Notably, unroofing of the upper Cenozoic strata in the center of the DFS and JS anticlines took place after ~3.2 Ma and ~1.6 Ma, respectively. This is consistent with seismic profiles, in which the youngest strata in the hinge of the DFS anticline are exclusively of Pliocene age (Fig. 7A), while those in the JS anticline are of Quaternary age (Fig. 7B). The composite paleomagnetic data show that the lithofacies with gray mudstone and halite, representing shallow lake to saline paleo-lake environment, were deposited in the two limbs (as the positions of the SG-3 and SG-1 drill cores) of the anticline (Figs. 6C, E and 7) yet deposition of this facies at the anticline hinge had stopped, base on inferred magnetostratigraphic ages (of the SG-2 and SG-1b drill cores) (Fig. 6B, D). Although previous deposition can be eroded in later stage, the multiple shifts of the basin depocenter due to tectonic uplift most likely resulted in the differential ages of the anticline centers (Zhang et al., 2014; Zhou et al., 2006). The anticline axes in the western Qaidam Basin has been active since the late Miocene, which were proved by the crustal shortening and basin clockwise rotation (Zhou et al., 2006; Bao et al., 2017; Li et al., 2017, 2018). According to the models for the development of the typical fault-bend fold (Suppe, 1983; Suppe et al., 1992; Poblet et al., 1997), Initial development of offlap growth strata are proposed at ~1.6 Ma and 3.2 Ma in the JS and DFS anticlines, indicating that the depositional histories of fold limbs were controlled by the transition between overlap and offlap growth at ~1.6 Ma and 3.2 Ma, respectively, in the centers of the JS and DFS anticlines (Fig. 7).

Seismic profiles and lithologic characteristics show that the thicker lacustrine sediments deposited on the forelimb in the western Qaidam Basin (Figs. 6 and 7), indicating different sedimentation rates in the limbs and in the anticline hinge over the same period of deposition (Fig. 6F). Stratigraphic thickness variations can be alternatively attributed to some factors, i.e. differential erosion from the different locations and the shift of the depocenter, but we suggest that the stratigraphic thickness of syntectonic growth strata in the western Qaidam Basin were mainly influenced by anticline development (Zhang et al., 2014), which was confirmed by seismic profiles crossing the Dafengshan and Jianshan anticlines indicating the transition between onlap and offlap since the late Pliocene (Fig. 7A). Therefore, the changes in the sedimentation rate (R_s) and uplift rate (R_u) in the shortened region can be quantitatively constrained by the growth strata (Suppe et al., 1992; Anastasio et al., 1997). It should be noted that the R_u is calculated by the formulae from the difference value of the stratigraphic thicknesses divided by the corresponding ages (Zhang et al., 2014), based on the strata within 0.1 Ma - 3.2 Ma from magnetostratigraphic ages of the four drill cores. Using the overlap and offlap depositional mechanisms described for the limb rotation model (Suppe, 1983; Poblet et al., 1997), the values of R_u of the drill cores SG-2 and SG-1b are 49.5 cm/ka and 35.1 cm/ka, as deduced from ages of principal magneto-zones of the four drill cores (as SG-1, SG-1b, SG-2, and SG-3) (Figs. 6 and 7C). These two large values of R_u for the DFS anticline and the JS anticline, together with the transition between overlap and offlap growth strata, define two important rapid stages of uplift associated with abruptly increased crustal shortening rate in the western Qaidam Basin, comparing with crustal shortening during the Miocene (Zhou et al., 2006), occurred at ~3.2 Ma and ~1.6 Ma.

Thus, 3.2 Ma and 1.6 Ma for youngest ages of the DFS and JS anticlines, as inferred by the combined magnetostratigraphic data not only represent the termination of depositions and the ending time of syntectonic overlap growth strata at the centers of the DFS and JS anticlines, but also document the onset of deposition of syntectonic offlap growth strata in the limbs of these two anticlines. Offlap growth strata

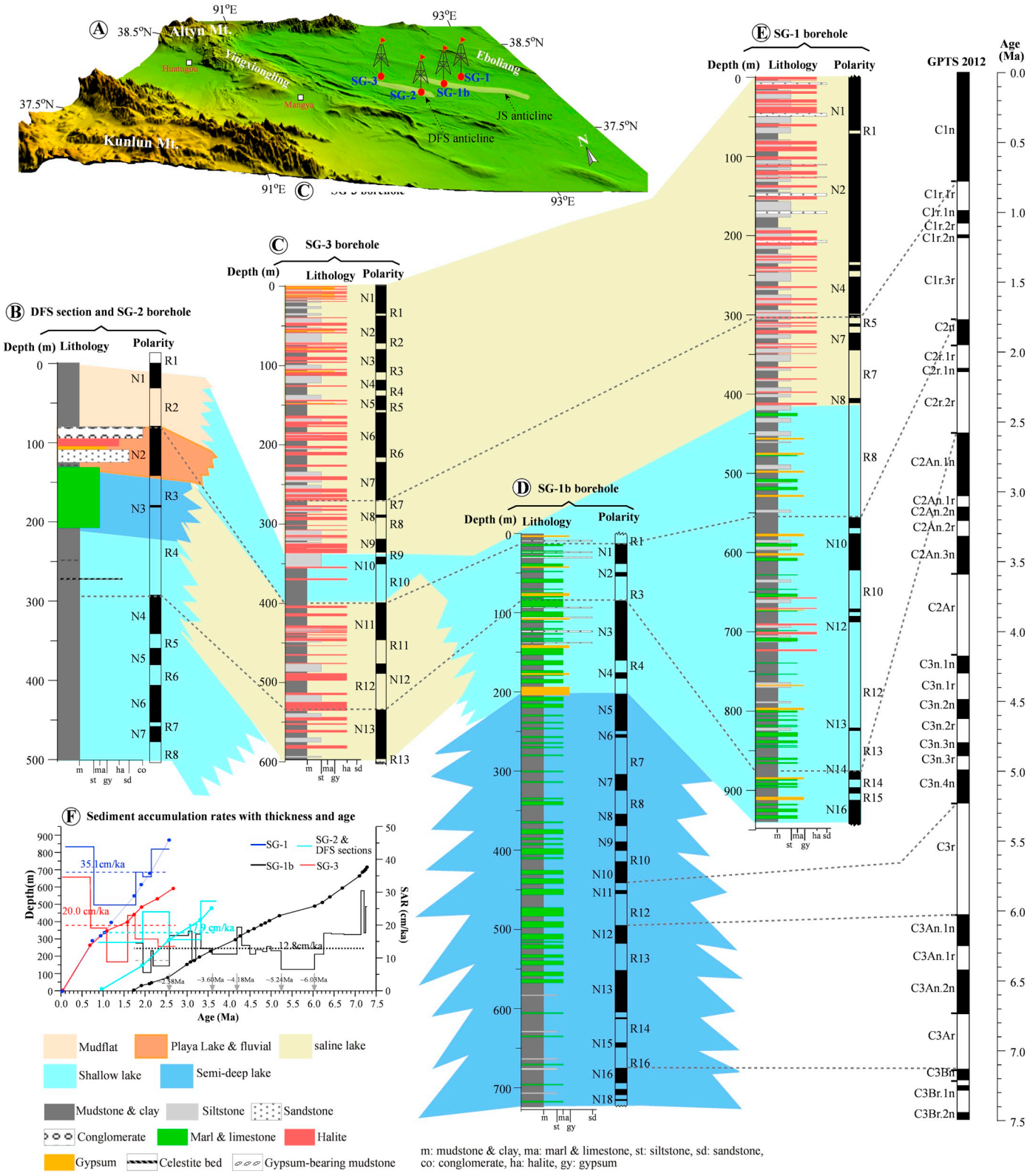


Fig. 6. DEM of the western Qaidam and the surrounding mountains showing the sites of the SG-1, SG-1b, SG-2 and SG-3 drill cores in the center of the western Qaidam Basin (A), the correlations of magnetic polarity sequences, lithologies and sedimentary facies (Zhang et al., 2012a, 2014; Wang et al., 2012; Cai et al., 2012) (B-E), and sediment accumulation rates (SAR) (Zhang et al., 2012a, 2014; Cai et al., 2012) with thickness – age plot based on the correlation of the polarity sequence with the GPTS 2012 (Gradstein et al., 2012) (F).

in the anticlines, developed at 3.2 Ma in the DFS ancticline and 1.6 Ma in the JS anticline in the center of the western Qaidam Basin, and indicate progressive younging of the off growth strata from SE to NW in horizontal space, caused by strong strain propagation in the western Qaidam Basin and rapid uplift of the NE Tibetan Plateau at 3.2 Ma and 1.6 Ma.

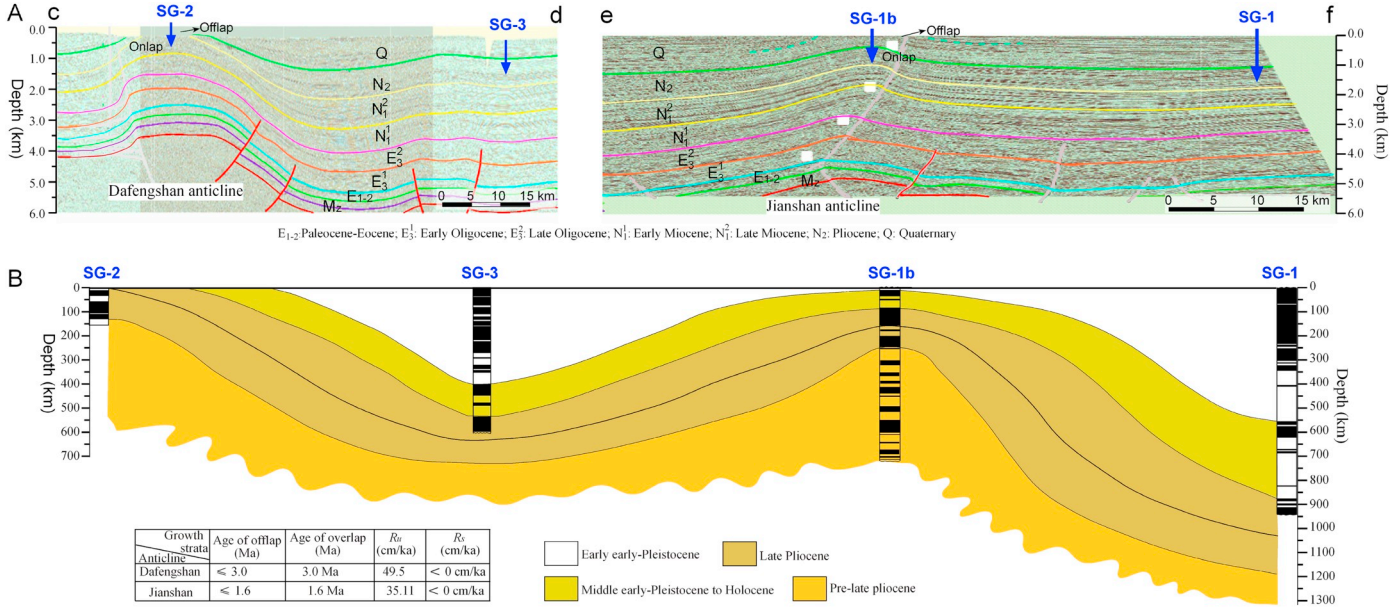


Fig. 7. (A) Seismic profiles along c-d and e-f shown in Fig. 1C, indicating that offlap growth strata of the JS anticline are younger than that of the DFS anticline, which developed in the late Pliocene and early Pleistocene, respectively. (B) Growth strata correlations between the SG-1, SG-1b, SG-2 and SG-3 drill cores according to their magnetostratigraphic results based on Zhang et al. (2014). Occurrence of offlap growth of the decollement are indicated at 1.6 Ma in the JS anticline and at 3.2 Ma in the DFS anticline. Uplift rates (R_u) and sediment accumulation rates (R_s) are calculated using the approach of Suppe (1983) and Poblet et al. (1997).

6.2. Driving mechanism of syntectonic growth strata

With the successive underthrusting of the India Plate into Asia, the entire Tibetan Plateau was subjected to intense south-north deformation during the late Cenozoic, leading to crustal shortening of Cenozoic strata within Cenozoic basins, tectonic uplift of surrounding mountains and rapid strike-slip displacement along large-scale faults (Harrison et al., 1995; Hurtado Jr et al., 2001; Zhou et al., 2006; Li et al., 2018; Bao et al., 2017; Zhang et al., 2018). Dynamic mechanisms for the south-west compression may have involved the deposition and deformation of syntectonic growth strata in the interior of the western Qaidam Basin (Zhou et al., 2006; Zhang et al., 2014; Bao et al., 2017). Deformation of syntectonic growth strata was accompanied by the shift of depocenters, the variation in sedimentary facies and the change of paleo-lake water level in the basin (Zhang et al., 2013, 2014). Comprehensive lithofacies and sedimentologic comparisons of the four drill cores SG-1, SG-1b, SG-2, and SG-3 indicate that the western Qaidam Basin had experienced two major uplift stages of syntectonic growth strata at ~3.2 Ma and ~1.6 Ma since the Pliocene (Fig. 8), directly related to rapid tectonic uplift of the NE Tibetan Plateau and strong basinward propagation of the thrust faults at the southern margin of the Qaidam Basin.

Firstly, strata in the limbs and hinges of the DFS and JS anticlines before 3.2 Ma are characterized by gray mudstone and interbeded mudstone and marl (Fig. 6B, C, D, E), indicating a depositional environment from a shallow lake to semi-deep lake in the interior of the western Qaidam Basin. At ~3.2 Ma, the hinge area of the DFS anticline began to be exposed above the lake surface, with the termination of sediment deposition (Fig. 7A). The SE limb of the DFS anticline and the syncline between the DFS anticline and the JS anticline still received lacustrine deposits including halite, mudstone and marl, as shown by the lithofacies from the DFS-Iand DFS-II outcrop sections and the SG-3 drill core (Figs. 6B, C and 8A). The observed dips of growth strata on the limbs also show large progressive changes caused by uplift of the core of the anticline at this time (Fig. 7). Enhanced tectonic uplift at ~3.2 Ma is indicated by a northward propagation or a synchronous folding of the DFS anticline due to northward displacement related to slip along the east Kunlun fault (Fig. 8A). Large crustal shortening occurred in the western Qaidam Basin, and the depocenter started to shift eastwards at ~3.2 Ma (Huang et al., 1996; Bao et al., 2017; Zhou et al., 2006; Fang et al., 2007). Crustal shortening and onset of the depocenter shift show that the western Qaidam Basin was rapidly compressed and shrunk at 3.2 Ma. Meanwhile, the depositional margins of the western Qaidam Basin started to receive coarse detritus with a rapid sedimentation rate since ~3.2 Ma as a result of sudden tectonic movement along the Altyn Tagh fault and the east Kunlun fault (Fang et al., 2007; Zhang et al., 2013; Zhang et al., 2018). Due to enhanced tectonic deformation of the Tibetan Plateau at ~3.2 Ma, the sedimentary environment responded with a significant change from lacustrine facies to fluvial facies in the rift basins of the northern Himalaya Mountains (Wang et al., 2008b; Zhang et al., 2020), the formation of the Kunlun Pass Basin in the middle section of the east Kunlun Shan (Mts.) (Yin et al., 1999; Song et al., 2005), the development of syntectonic angular unconformities in the adjacent Hexi corridor, Guide Basin and the Tarim Basin around the Qaidam Basin (Fang et al., 2005; Sun et al., 2010), the widespread deposition of thick piedmont conglomerates in the Cenozoic basins at the periphery of the Tibetan Plateau (Burbank and Johnson, 1982; Zhong and Ding, 1996; Zheng et al., 2000; Song et al., 2001; Fang et al., 2005; Li et al., 2014; Zhang et al., 2017), and the inception of aeolian loess-red clay deposition in the western Chinese Loess Plateau (Zan et al., 2016, 2018).

Secondly, the inferred magnetostratigraphic ages propose that termination of sediment deposition in the core of the JS anticline occurred at ~1.6 Ma, inferred by the youngest age of the SG-1b drill core (Zhang et al., 2014), indicating that offlap growth strata started to develop in the JS anticline since ~1.6 Ma. About 40% of the total crustal shortening after 1.6 Ma, demonstrated by balanced cross-sections in the western Qaidam Basin (Zhou et al., 2006; Wang et al., 2010; Bao et al., 2017), reveals that the Tibetan Plateau emerged an active stage of shortening at ~1.6 Ma, caused by the ongoing convergence between India and Asia (Fig. 8B). Major shortening basinwards of the eastern Kunlun fault led to the continuation of sandstone and conglomerate deposition in the SE limb of the DFS anticline (Figs. 6B; 8B), gradual shallowing of lake water with dominant halite in the NW limb of the DFS and both limbs of the JS anticlines (Fig. 6C, E), and the formation of playa lakes in the region near the basin margin along the east Kunlun Mts. (Fig. 8B). Shortening at this stage was associated with the

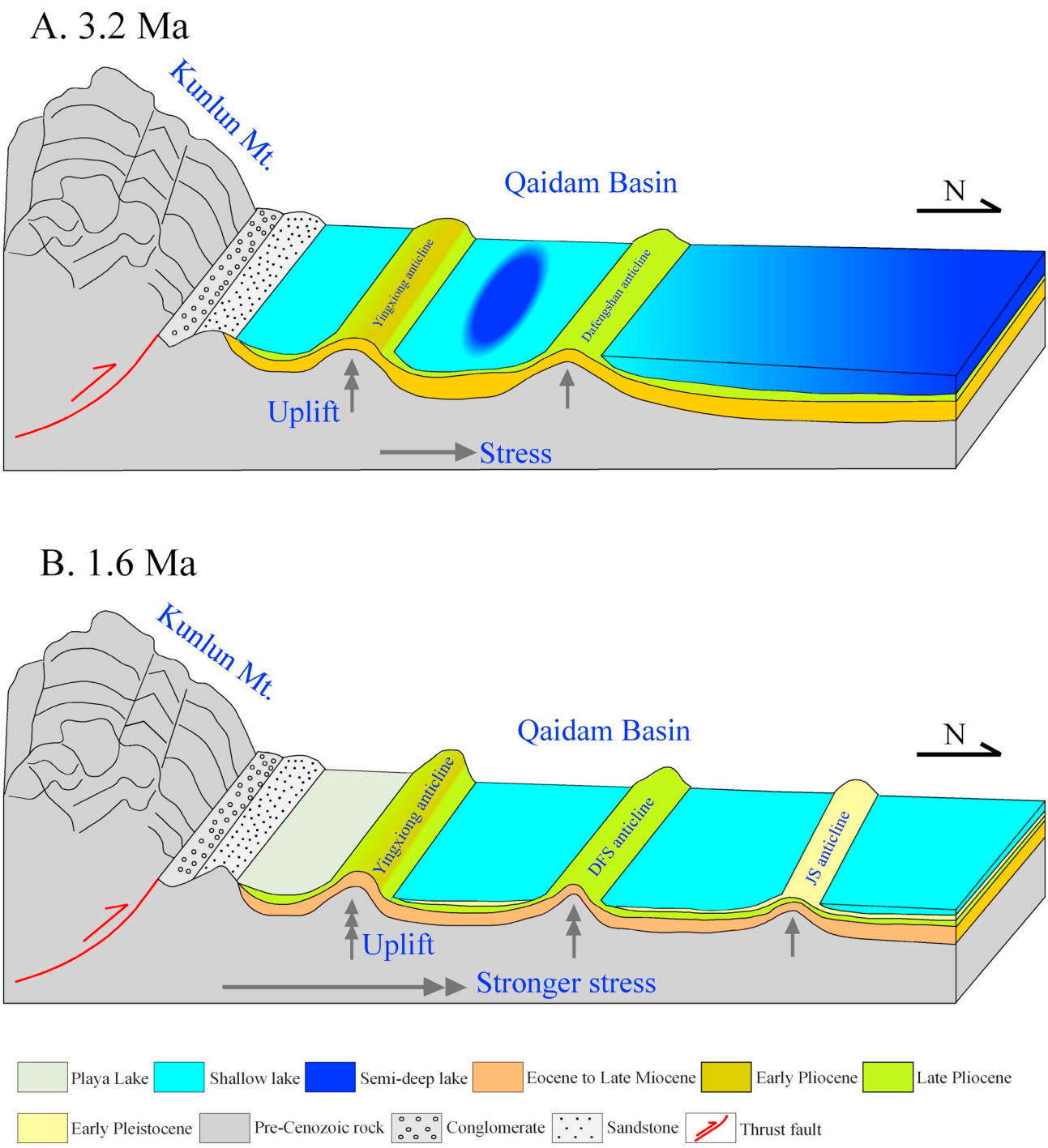


Fig. 8. Dynamic mechanisms of syntectonic growth strata in the western Qaidam Basin illustrating the relationship between strong basinward propagation of the east Kunlun thrust fault and spatiotemporal variation of sedimentary characteristics in the interior of the western Qaidam Basin at 3.2 Ma (A) and 1.6 Ma (B) since the late Pliocene.

occurrence of glaciogenic diamictite in the Cenozoic basins from the southern Tibetan Plateau (Wang et al., 1996; He et al., 2016), rapid uplift with an average rate of 5 mm/a of the western Himalaya syntaxis (Zeitler, 1982), and accumulation of boulder conglomerates in the southern Himalaya molasse (Rösler et al., 1997; Mugniera et al., 1999; Sakai et al., 2006), corresponding to “phase C” of the Qingzang movement at 1.6 Ma (Li et al., 1996; Li and Fang, 1999).

After 1.6 Ma, syntectonic offlap growth strata became more prominent, and the basin depocenter shifted eastwards and ultimately withdrew from the western Qaidam Basin toward the Qaerhan region at 0.1 Ma (Shen et al., 1993; Zhang et al., 2014). The termination of evaporite deposition between the limbs from the DFS and JS anticlines occurred at ~0.1 Ma (Figs. 6C, E and 7B). Since ~0.1 Ma, the western Qaidam Basin has been subjected to unroofing erosion by the strong Westerlies and the regional wind at our study region, and finally formed the present geomorphic features that define the Yardang landform (Wang et al., 2007; Kapp et al., 2011).

7. Conclusions

Magnetic polarity stratigraphic analysis of the stratigraphic sequence from the Dafengshan (DFS) sections and the SG-2 borehole in

the DFS anticline in the western Qaidam Basin yields estimated ages between 3.7 Ma and 0.9 Ma. The youngest age of the core of the DFS anticline is dated at ~3.2 Ma, indicating that syntectonic offlap growth strata in the western Qaidam Basin developed before ~3.2 Ma. A comparison with magnetostratigraphic results from nearby drill cores located in the western Qaidam Basin demonstrates that syntectonic offlap growth strata progressively become younger from southwest to northeast in the center of the western Qaidam Basin since the Pliocene. We suggest that the development of syntectonic growth strata in the anticlines from the Qaidam Basin was mainly driven by strong basinward propagation of the east Kunlun fault and pulses of rapid tectonism of the NE Tibetan Plateau.

CRediT authorship contribution statement

Weilin Zhang: Supervision, Writing - original draft, Writing - review & editing.**Erwin Appel:** Supervision, Writing - original draft, Writing - review & editing.**Xiaomin Fang:** Supervision, Writing - original draft, Writing - review & editing.**Fabian Setzer:** Supervision, Writing - original draft, Writing - review & editing.**Chunhui Song:** Supervision, Writing - original draft, Writing - review & editing.**Qingquan Meng:** Supervision, Writing - original draft, Writing - review & editing.**Maodu Yan:** Supervision, Writing - original draft, Writing - review & editing.

Declaration of competing interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tecto.2020.228401.

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